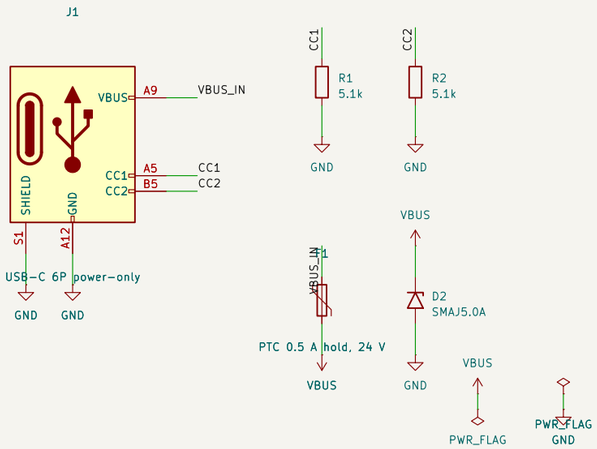
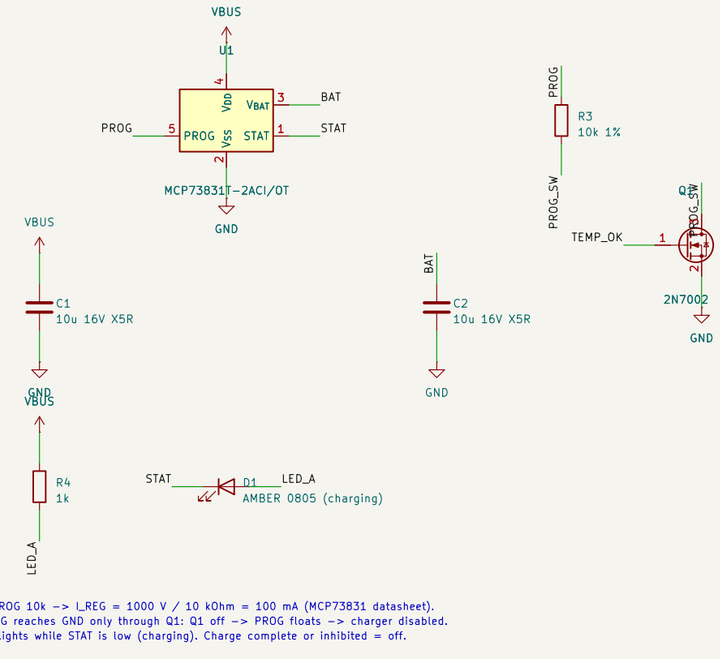
NIVA CHARGING DOCK — ENGINEERING PROTOTYPE — NOT FOR FABRICATION

01 USB-C SINK (5 V ONLY)



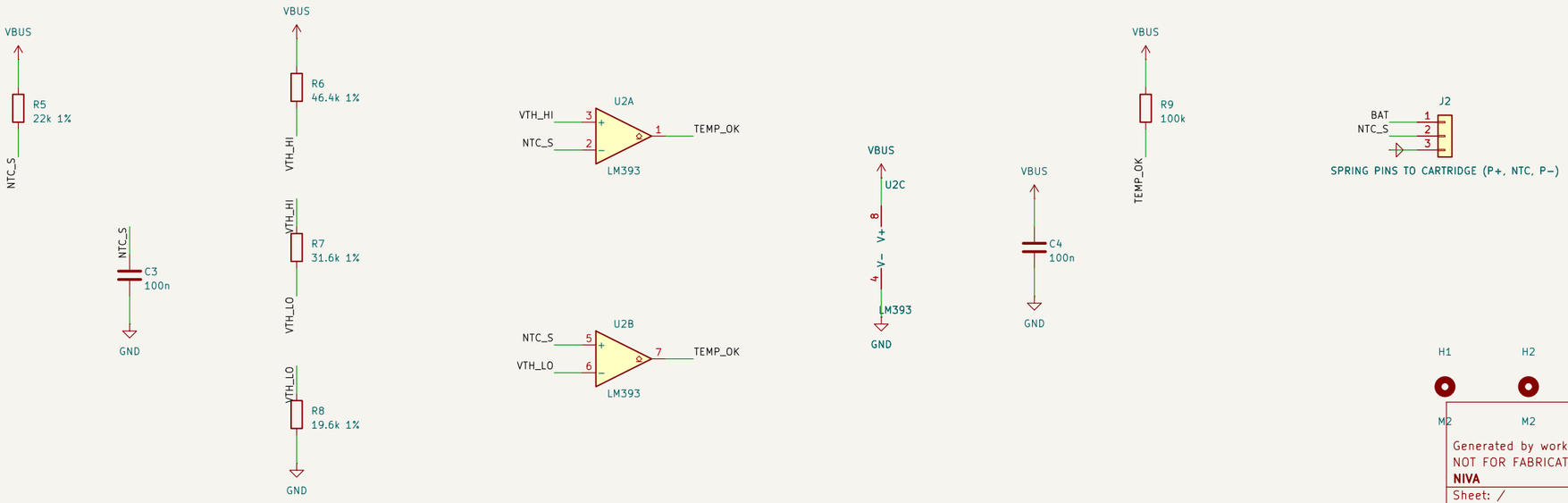
D2 TVS (Bourns SMAJ5.0A) after F1, VBUS to GND, as in MCP73831 DS20001984H 6.1.1.2: clamps hot-plug ringing (VBR 6.40–7.00 V at 10 mA vs the 7.0 V VDD abs. max). A sustained overvoltage makes D2 conduct and trips F1 (24 V PTC).

02 LINEAR CHARGER 4.20 V / 100 mA

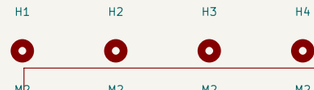


R_PROG 10k -> I_REG = 1000 V / 10 kOhm = 100 mA (MCP73831 datasheet).
PROG reaches GND only through Q1: Q1 off -> PROG floats -> charger disabled.
D1 lights while STAT is low (charging). Charge complete or inhibited = off.

03 PACK NTC WINDOW (HARDWARE, NO FIRMWARE)



NTC_S = VBUS x R_NTC / (22k + R_NTC). Window (ratio of VBUS): cold/open trip 0.525 (VTH_HI), hot/short trip 0.201 (VTH_LO).
10k B3435 NTC: charges 3.7..41.2 C nominal, 2.2..43.0 C worst case (1 % R and NTC, LM393 9 mV / 400 nA). B3950: 6.3..39.0 C.
No cartridge: NTC_S = VBUS (5 V) -> unit A pulls TEMP_OK low -> no charge. Only VTH_HI (2.62 V) must be inside the LM393 common-mode range (VCC - 1.5 V at 25 C, VCC - 2 V over temperature): an input outside it still gives a correct output.
No hysteresis. TEMP_OK->VTH_HI feedback would fix the cold edge but makes the hot edge unstable (shared divider): Rev C.3.



Generated by work/build_niva_schematic_revC.py
NOT FOR FABRICATION. Values, protection and layout require bench validation.
NIVA

Sheet: /
File: NIVA-dock.kicad_sch

Title: NIVA cartridge charging dock | USB-C, MCP73831, hardware NTC windo

Size: A3	Date: 2026-09-28	**Rev: C — ENGINEERING PROTO**
KiCad E.D.A. 10.0.7		Id: 1/1